A DEVELOPER'S GUIDE

LangChain, LlamaIndex and CrewAI

What, How, When ?

All three are LLM orchestration frameworks. They connect models to your data, tools and workflows. They overlap, and choosing between them is easier once you see how each one runs.

swipe →

Each one is built around something different

LangChain

`built around: steps & state`

A toolkit for wiring models, tools and memory together. LangGraph, its lower-level runtime, runs agents as a graph you control.

LlamaIndex

`built around: documents & queries`

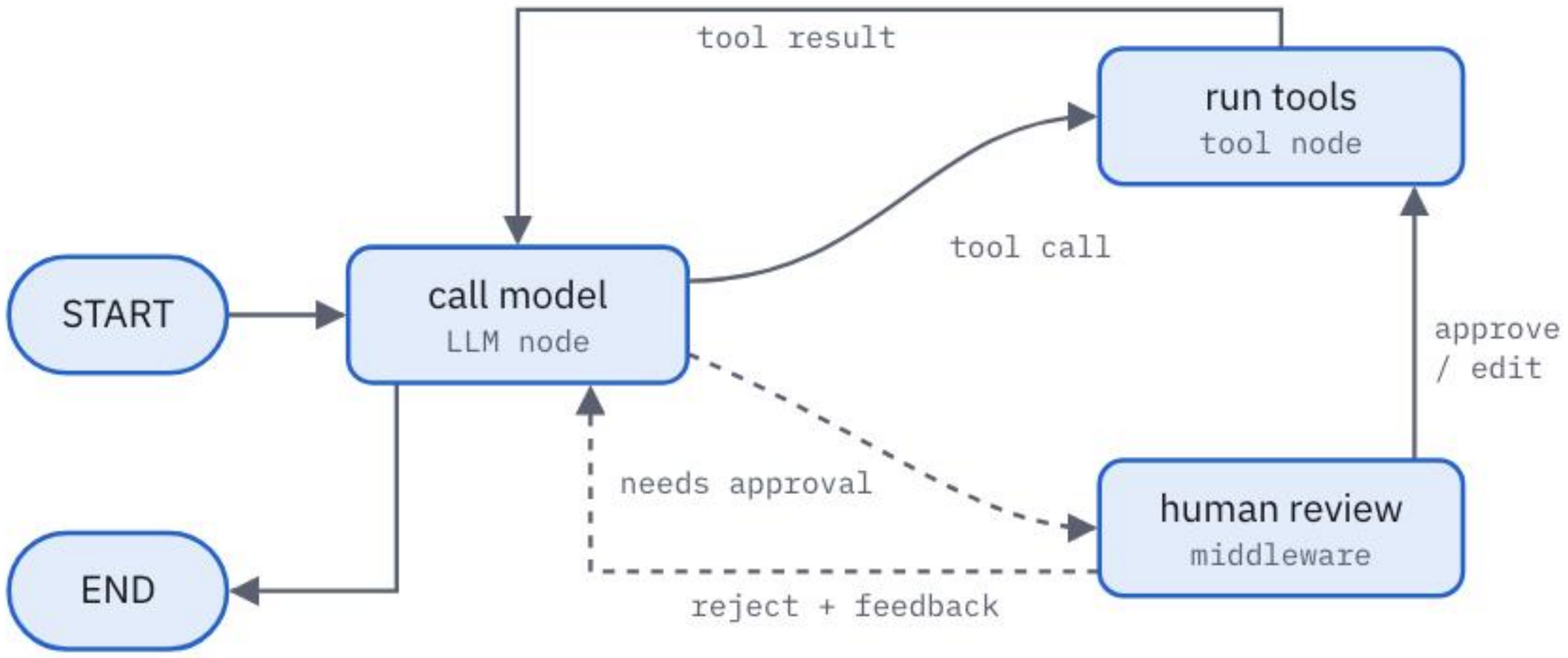
A data framework for getting your own documents into an LLM: load, parse, index, retrieve, answer with sources.

CrewAI

`built around: roles & tasks`

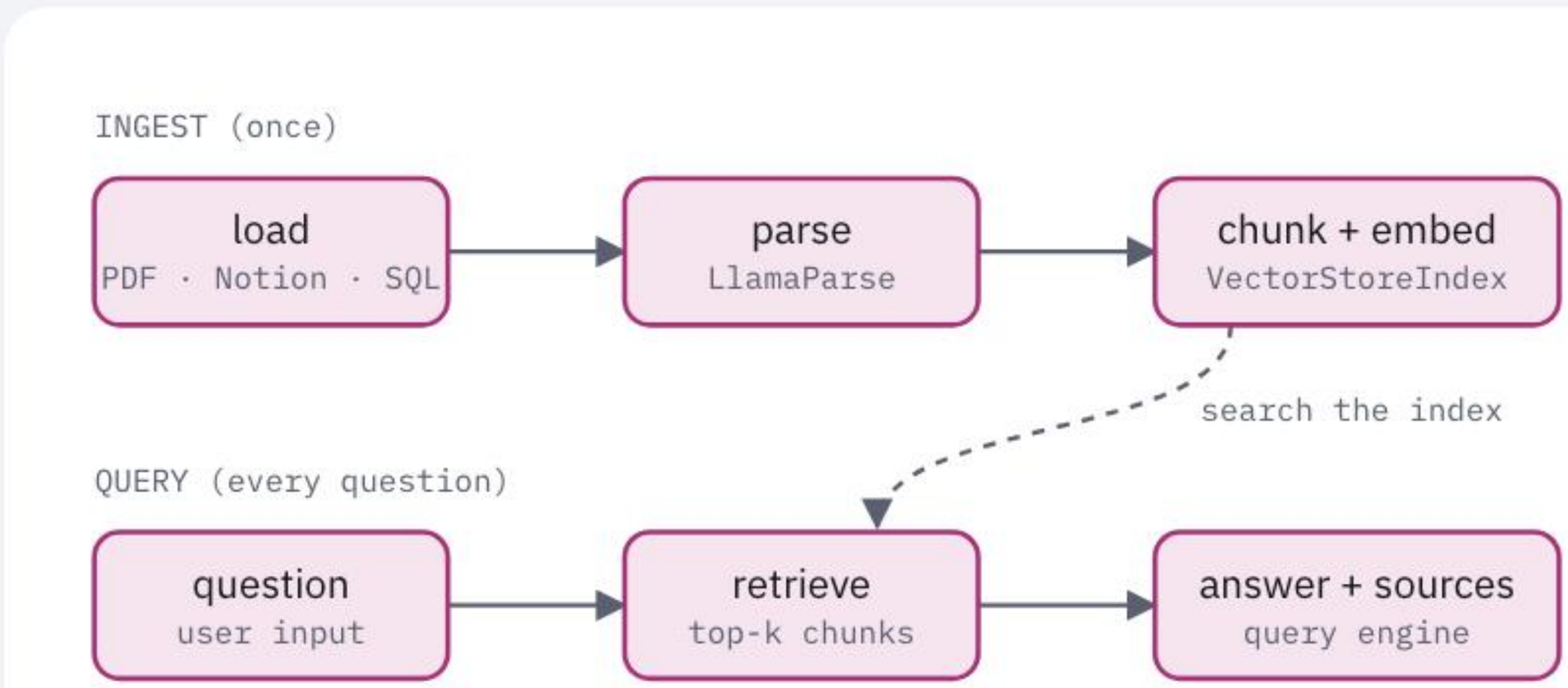
A multi-agent framework where you describe a team, give them tasks, and let the crew work through them.

An agent is a loop in a graph



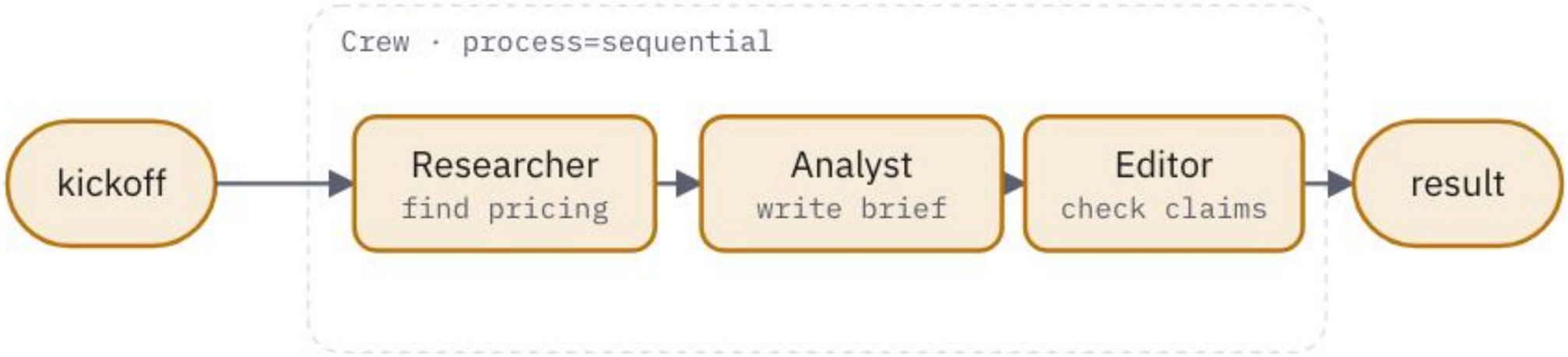
- The model decides whether to call a tool or answer.
- Tool results always go back to the model. Only the model can end the loop.
- Optional human review pauses before a tool runs: approve, edit or reject.

Ingest once, retrieve on every question



- Load your data, parse messy files, then chunk, embed and index it.
- For each question, the retriever finds the most relevant chunks.
- The model answers from those chunks and can cite them as sources.

Describe a team, give it tasks



- Each agent has a role, a goal and a backstory.
- Each task says what output is expected, and can use earlier tasks as context.
- Tasks run in order, or a manager agent delegates them (hierarchical mode).

WHEN TO USE WHICH ?

Which one should you reach for ?

Chat with your company's PDFs and docs

- **LlamaIndex**

An agent with branches, retries and human approval

- **LangChain / LangGraph**

A team of agents: researcher, writer, reviewer

- **CrewAI**

Extract fields from messy invoices and forms

- **LlamaIndex**

A quick multi-step demo for tomorrow

- **CrewAI**

A production agent you need to debug and monitor

- **LangChain / LangGraph**

In real projects people sometimes mix them. A LlamaIndex retriever as a tool inside a LangGraph or CrewAI agent is a common setup.

HOW THEY DIFFER

The differences at a glance

CORE ABSTRACTION

LangChain
Agent loop built on a graph of nodes, edges and shared state

LlamaIndex
Index, retriever and query engine, plus event-driven Workflows

CrewAI
Agents with a role, goal and backstory, Tasks, and a Crew

CONTROL FLOW

LangChain
You define it explicitly

LlamaIndex
Mostly a pipeline. Workflows add steps and events

CrewAI
Sequential, hierarchical, or Flows for explicit control

STRONGEST AT

LangChain
Custom, stateful agents with full control

LlamaIndex
RAG and document-heavy apps

CrewAI
Getting a team of role-based agents working fast

TRADE-OFF

LangChain
More concepts to learn, and you design the graph yourself

LlamaIndex
Less focused on multi-agent orchestration

CrewAI
Less step-by-step control unless you use Flows

HOW THEY DIFFER

What does it cost ?

FRAMEWORK COST

LangChain

Free, open source (MIT)

LlamaIndex

Free, open source (MIT)

CrewAI

Free, open source (MIT)

WHAT YOU PAY FOR

LangChain

LLM calls. One per pass through the loop, so an agent using 3 tools makes at least 4 calls

LlamaIndex

LLM calls. Embeddings once when you index, then about one call per question

CrewAI

LLM calls. At least one run per agent, plus the manager's calls in hierarchical mode

PAID PLATFORM (AS OF SEPT 2026)

LangChain

LangSmith: free for 1 seat and 5k traces a month. Plus is $39 per seat a month

LlamaIndex

LlamaParse: 10k free credits a month. Starter $50 a month, Pro $500 a month. 1,000 credits = $1.25

CrewAI

CrewAI platform: 50 free workflow runs a month. Enterprise has custom pricing

The frameworks are free. Your bill depends mostly on the model you choose and how many calls your design makes.

REFERENCES

Where to read more

PAPERS

1. Rasheed et al. (2026). Agentic Frameworks for Reasoning Tasks: An Empirical Study.
   arxiv.org/abs/2604.16646
2. Ferrag, Tihanyi and Debbah (2025). From LLM Reasoning to Autonomous AI Agents: A Comprehensive Review.
   arxiv.org/abs/2504.19678
3. Singh et al. (2025). Agentic Retrieval-Augmented Generation: A Survey on Agentic RAG.
   arxiv.org/abs/2501.09136

DOCUMENTATION

1. LangChain docs.
   docs.langchain.com
2. LangChain and LangGraph 1.0 release.
   langchain.com/blog/langchain-langgraph-1dot0
3. LlamaIndex docs.
   developers.llamaindex.ai
4. CrewAI docs.
   docs.crewai.com

PRICING

1. LangSmith pricing.
   langchain.com/pricing
2. LlamaParse and LlamaCloud pricing.
   llamaindex.ai/pricing
3. CrewAI pricing.
   crewai.com/pricing